Design a low-power **64-bit RISC-V compute subsystem** for **mobile XR**
under a **3 W, 3 nm-class LP envelope**,
returning a **DSE report with evidence and rejected alternatives**.

Intent

Improve mobile XR efficiency under power, reliability, memory, and deployment constraints.

Task

Bounded DSE for a RISC-V compute subsystem, starting with an accelerator and memory slice.

Design space

Legal accelerator, memory, and vector choices; invalid power violations and deferred software work.

Representation

XRBench traces, architecture parameters, compiler assumptions, power model, and uncertainties.

Environment

Simulator or cost model plus workload harness; actions change tiling, memory, or vector width.

Method role

Generate candidates, predict cost, optimize, critique assumptions, and write evidence reports.

Feedback budget

Many cheap proxies, fewer simulations, and rare high-fidelity checks before review.

Evidence

Pareto comparison over latency, energy, area proxy, memory traffic, QoS, and sensitivity.

Negative traces

Power violations, missed real-time bounds, bandwidth overruns, unsupported software, or proxy mismatches.

Rejection authority

Declared checks, constraint violations, QoS misses, cross-tool disagreement and independent review.

Commitment boundary

Authorize a higher-fidelity study only; no RTL, integration, or product commitment.

Human decision

The architect may refine, escalate fidelity, change scope, or reject the candidate.

Example card, not a completed design claim.